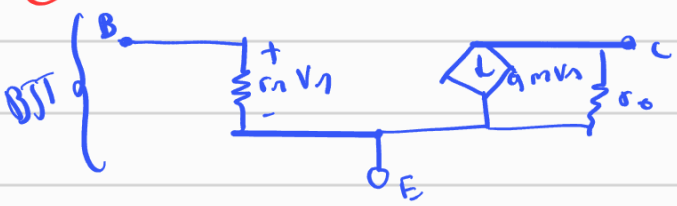
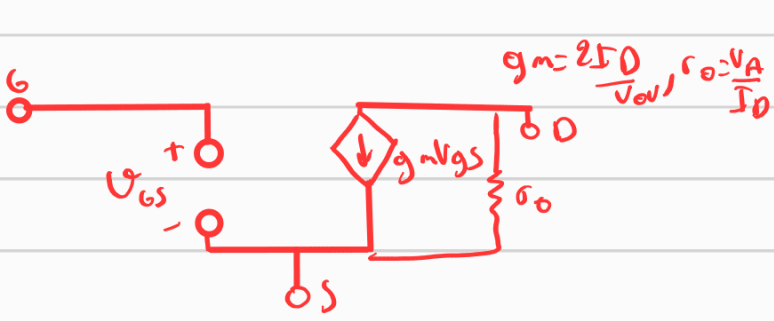
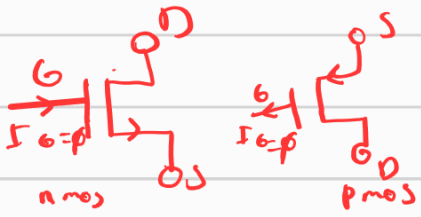
Mosfet small signal





$g_m = \frac{2I_D}{V_{OV}}$, $r_o = \frac{V_A}{I_D}$



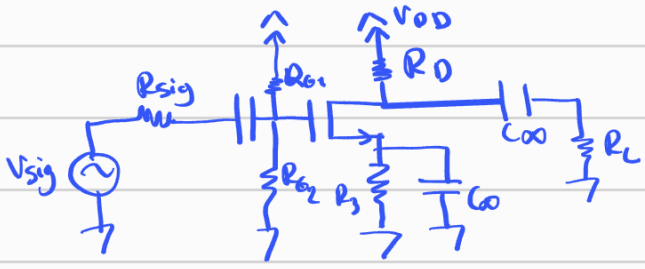
- Common Source emitter
- Common Source " R_S emitter with R_E
- Common Drain Collector (Source Follow)
- Common - Gate Base

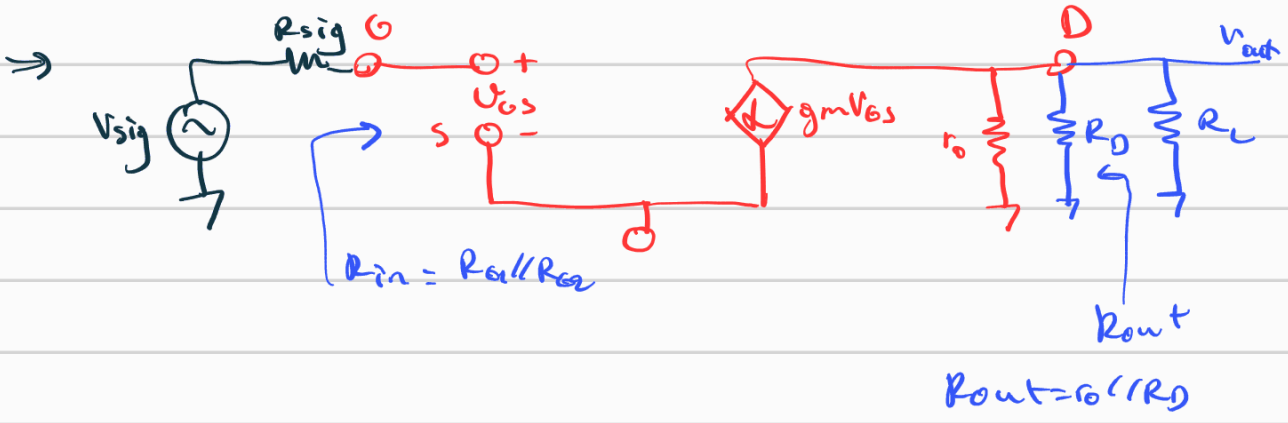
Base → Gate
Emitter → Source
Collector → Drain

Common source Amplifier

- Voltage Amp (High)
- $R_{in} = \infty$ } Inf
- $R_{out} = r_o // R_D // R_L$ } High

R_{out} (intrinsing) $= r_o$





$R_{out} = r_o // R_D$

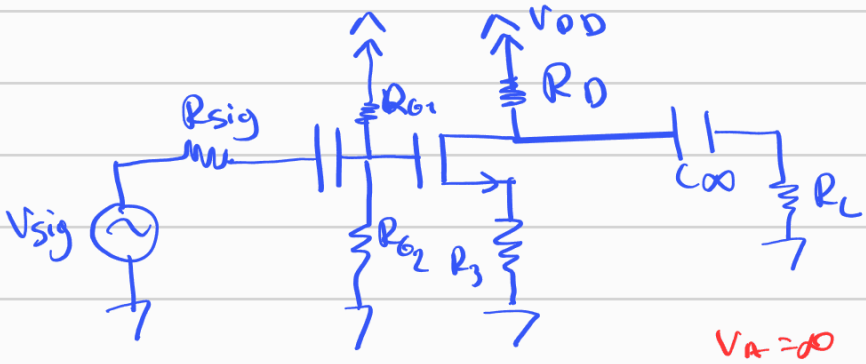
$$A_v = \frac{v_{out}}{v_{sig}} = \frac{(r_o \,//\, R_D \,//\, R_L) \times -g_m v_{gs}}{v_{sig}}$$

$$v_{gs} = \frac{(R_{G1} // R_{G2})}{(R_{sig} + R_{G1} // R_{G2})}$$

$$= -(r_o // R_D // R_L)\, g_m \cdot \left(\frac{R_{G1} // R_{G2}}{R_{sig} + R_{G1} // R_{G2}}\right)$$

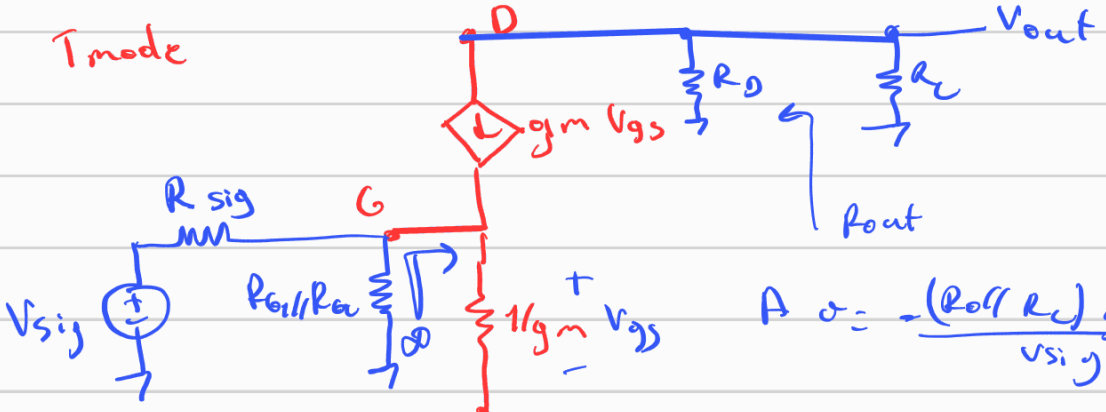
$R_{sig} \approx 0 \quad A_v = -g_m (R_D // R_L // r_o)$

Common Source with R_S



$V_A = \infty$ Channel length modulation ignored

T mode



$$A_v = \frac{-(R_o // R_L)\, g_m v_{gs}}{v_{sig}}$$

$$R_{in} = R_{G1} // R_{G2} // \infty = R_{G1} // R_{G2}$$

$$\Rightarrow A_v = \frac{-(R_o // R_L) g_m v_{gs}}{v_{sig}}$$

$R_{sig} \approx 0$

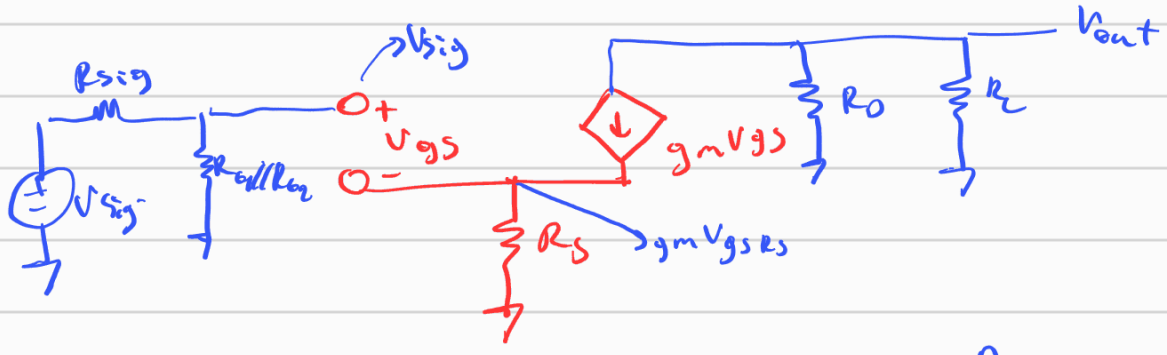
$$= \frac{-(R_o // R_L) g_m v_{gs}}{v_{gs}(1 + g_m R_S)}$$

$$\Rightarrow v_{sig} = v_{gs} + v_{gs} g_m \cdot R_S$$

$$v_{sig} = v_{gs}(1 + g_m R_S)$$

$$\approx A_v = \frac{-(R_o // R_L)}{R_S}$$

π model



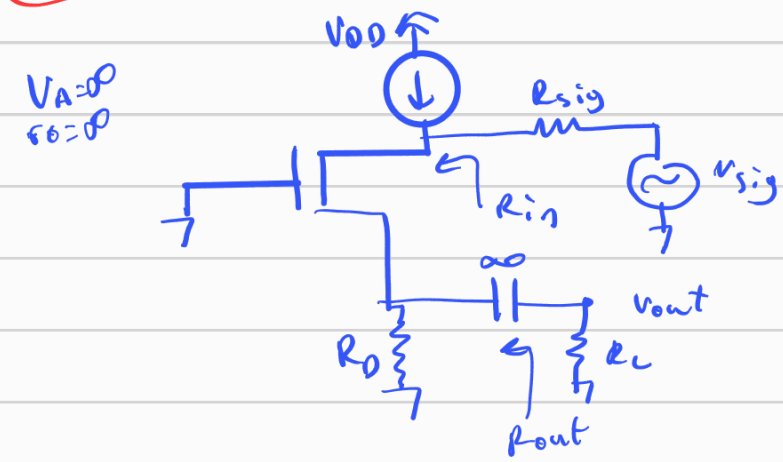
$V_{sig} = -g_m V_{gs} R_S$

$V_{sig} = (1 + g_m R_S) V_{gs}$

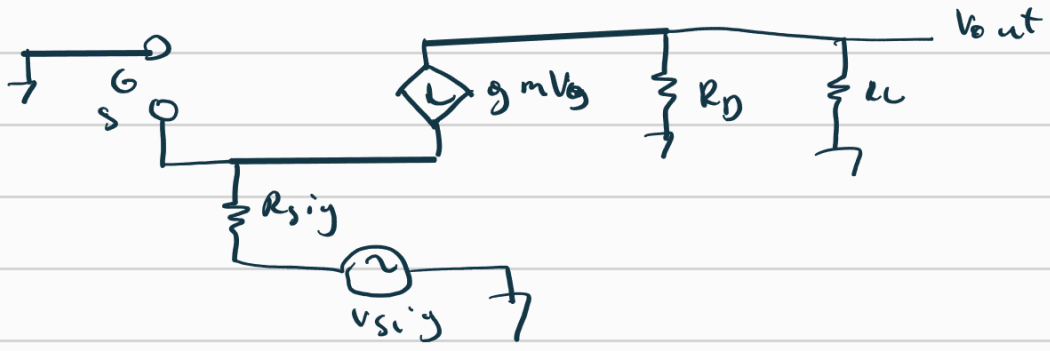
$A_v = \frac{V_{out}}{V_{sig}} = \frac{-g_m (R_D \| R_L) V_{gs}}{(1 + g_m R_S) \cdot V_{gs}}$

$A_v \cong -\frac{R_D \| R_L}{R_S}$ ($g_m R_S \gg$)

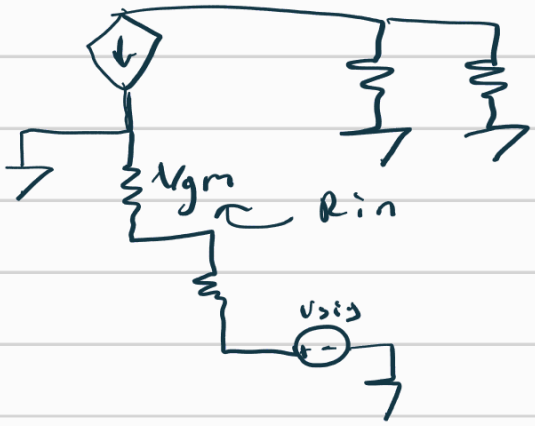
Common-Gate



T model örneği



T mode



$R_{in} \cong R_{gm}$ very small

$R_{out} = R_D$ moderate/high

$A_v \Rightarrow$ high

$R_{sig} = 0 \Rightarrow V_{sig} = -V_{gs}$

$V_{out} = -g_m (R_D \| R_L) V_{gs}$

$\Big\} A_v = g_m (R_D \| R_L)$

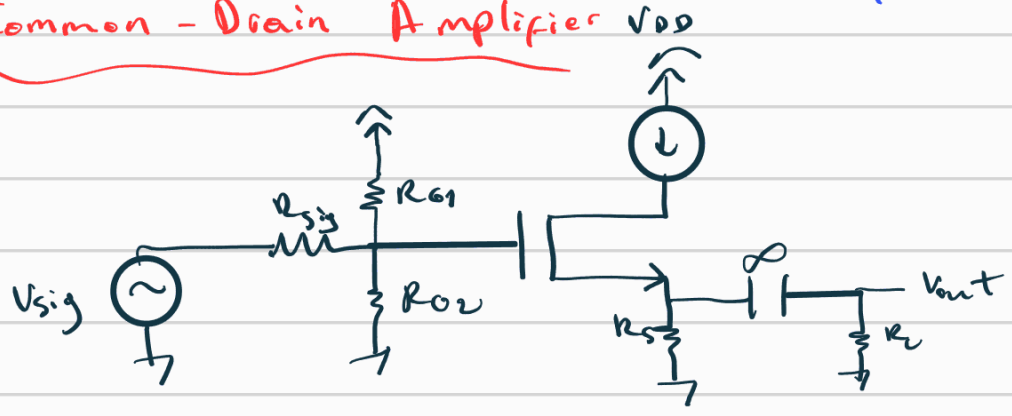
$R_{sig} \neq 0$

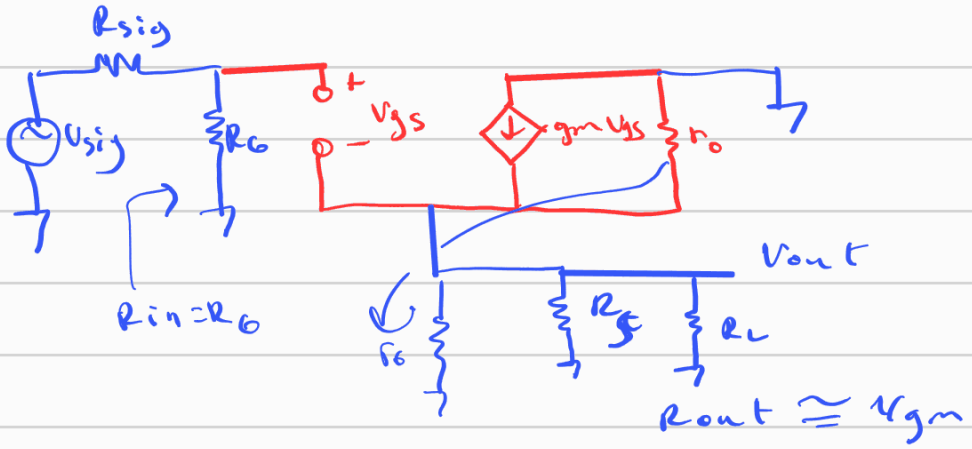
$V_{sig} = -g_m V_{gs} R_{sig} - V_{gs}$

$V_{out} = -g_m (R_D \parallel R_L) \cdot V_{gs}$

$A_v = \frac{V_{out}}{V_{sig}} =$

Common - Drain Amplifier (Source follower)





$R_{out} \cong \frac{1}{g_m}$

$V_{out} = g_m V_{gs} (R_S \parallel R_L \parallel r_o)$

$R_{sig} = 0$

$V_{gs} = V_{sig} - g_m V_{gs} (r_o \parallel R_L \parallel R_S)$

$V_{sig} = V_{gs} (1 + g_m (r_o \parallel R_L \parallel R_S))$

$\Rightarrow A_v = \frac{g_m V_{gs} (R_L')}{1 + g_m V_{gs} (R_L')}$

< 1